Customer Shopping Behavior Analysis

A comprehensive data-driven study of 3,900 customer transactions

Python | SQL (MySQL) | Power BI

3,900	18	3.75	$59.76
Customers	Features	Avg Rating	Avg Purchase

1. Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Attribute	Detail
Rows	3,900
Columns	18
Missing Data	37 values in Review Rating column

Key Features

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python, following a structured pipeline to ensure data quality before analysis.

Data Loading	Imported the dataset using **pandas**.
Initial Exploration	Used df.info() to check structure and .describe() for summary statistics.
Missing Data Handling	Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
Column Standardization	Renamed columns to *snake_case* for better readability and documentation.
Feature Engineering	Created **age_group** column by binning customer ages; created **purchase_frequency_days** column from purchase data.
Data Consistency Check	Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
Database Integration	Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Python Output – Data Exploration

The screenshots below show the dataset structure (**df.info()**) and descriptive statistics (**df.describe()**) generated during the initial exploration phase.

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

Figure 1 – df.info() output (dataset structure)

```
        Customer ID         Age Gender Item Purchased  Category  ... Discount Applied Promo Code Used Previous Purchases Payment Method Frequency of Purchases
count   3900.000000 3900.000000   3900           3900      3900  ...             3900            3900        3900.000000           3900                   3900
unique          NaN         NaN      2             25         4  ...                2               2                NaN              6                      7
top             NaN         NaN   Male         Blouse  Clothing  ...               No              No                NaN         PayPal         Every 3 Months
freq            NaN         NaN   2652            171      1737  ...             2223            2223                NaN            677                    584
mean    1950.500000   44.068462    NaN            NaN       NaN  ...              NaN             NaN          25.351538            NaN                    NaN
std     1125.977353   15.207589    NaN            NaN       NaN  ...              NaN             NaN          14.447125            NaN                    NaN
min        1.000000   18.000000    NaN            NaN       NaN  ...              NaN             NaN           1.000000            NaN                    NaN
25%      975.750000   31.000000    NaN            NaN       NaN  ...              NaN             NaN          13.000000            NaN                    NaN
50%     1950.500000   44.000000    NaN            NaN       NaN  ...              NaN             NaN          25.000000            NaN                    NaN
75%     2925.250000   57.000000    NaN            NaN       NaN  ...              NaN             NaN          38.000000            NaN                    NaN
max     3900.000000   70.000000    NaN            NaN       NaN  ...              NaN             NaN          50.000000            NaN                    NaN

[11 rows x 18 columns]
```

Figure 2 – df.describe() statistical summary

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in **MySQL** to answer ten key business questions. Each query result is shown below.

Q1 – Revenue by Gender

Compared total revenue generated by male vs. female customers.

gender	total_revenue
Male	157890
Female	75191

Q2 – High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount_usd
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67

customer_data 2 ×

Q3 – Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Q4 – Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	Average Purchased Amount
Express	60.48
Standard	58.46

Q5 – Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	average_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Q6 – Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Q7 – Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	total_customers
Loyal	3116
Returning	701
New	83

Q8 – Top 3 Products per Category

Listed the most purchased products within each product category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Q9 – Repeat Buyers & Subscriptions

Checked whether customers with more than 5 purchases are more likely to subscribe.

subscription_status	repeated_customers
Yes	958
No	2518

Q10 – Revenue by Age Group

Calculated total revenue contribution of each customer age group.

age_group	total_revenue
young_adult	62143
middle_age	59197
adult	55978
seniors	55763

5. Dashboard in Power BI

Finally, we built an **interactive dashboard in Power BI** to present insights visually. The dashboard includes filters for Subscription Status, Shipping Type, Category, and Discount, along with key KPI cards, charts for revenue by category, subscription breakdown, and a Revenue by Age Group chart segmented by Gender.

Figure 3 – Power BI Customer Behavior Dashboard

6. Business Recommendations

1	**Boost Subscriptions** Promote exclusive benefits and early-access offers for subscribers to grow the currently small subscriber base (7.3%).
2	**Customer Loyalty Programs** Reward repeat buyers with points or tiered benefits to move them into the Loyal segment and increase lifetime value.
3	**Review Discount Policy** Balance promotional discounts with margin control — high discount rates on items like Hats (50%) and Sneakers (~50%) may erode profitability.

4	**Product Positioning** Highlight top-rated products (Gloves, Sandals, Boots) and best-sellers (Jewelry, Blouse, Jacket) in marketing campaigns.
5	**Targeted Marketing** Focus efforts on high-revenue age groups (Young Adults) and express-shipping users who demonstrate higher average purchase amounts.

This report was generated from a data pipeline using Python, MySQL, and Power BI.